

Date: April 22, 2025

Physics Spring Regents

Name:

Class of:

Teacher:

Class period:

Part C

Answer all questions in this part.

Directions (66–85): Record your answers in the spaces provided in your answer booklet. Some questions may require the use of the 2006 Edition Reference Tables for Physical Setting/Physics.

Base your answers to questions 66 through 70 on the information below and on your knowledge of physics.

A car, initially traveling at a velocity of 25 meters per second west, accelerates uniformly at a rate of 1.5 meters per second squared east for 6.0 seconds.

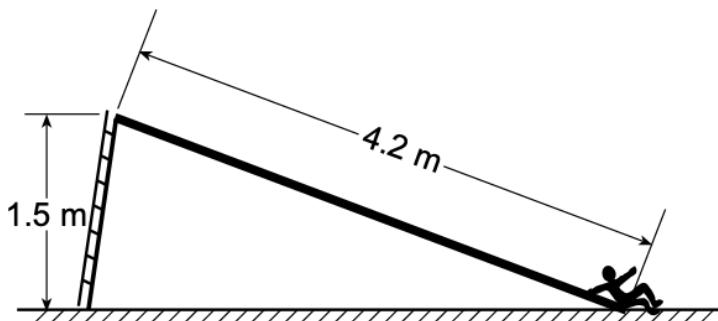
66–67. Calculate the magnitude of the car's velocity after the 6.0 seconds. *Show all work, including the equation and substitution with units.* [2]

68. Determine the direction of the car's velocity after the 6.0 seconds. [1]

69–70. Calculate the total distance traveled by the car during the 6.0 seconds. *Show all work, including the equation and substitution with units.* [2]

Base your answer to questions 71 through 76 on the information and diagram below and on your knowledge of physics.

Starting from rest at the top of the slide, a 40. kilogram child accelerates uniformly along the surface of a 4.2-meter-long slide at a rate of $0.30 \text{ meter per second squared}$ until she reaches the bottom of the slide.



71–72. Calculate the speed of the child at the instant she reaches the bottom of the slide. *Show all work, including the equation and substitution with units.* [2]

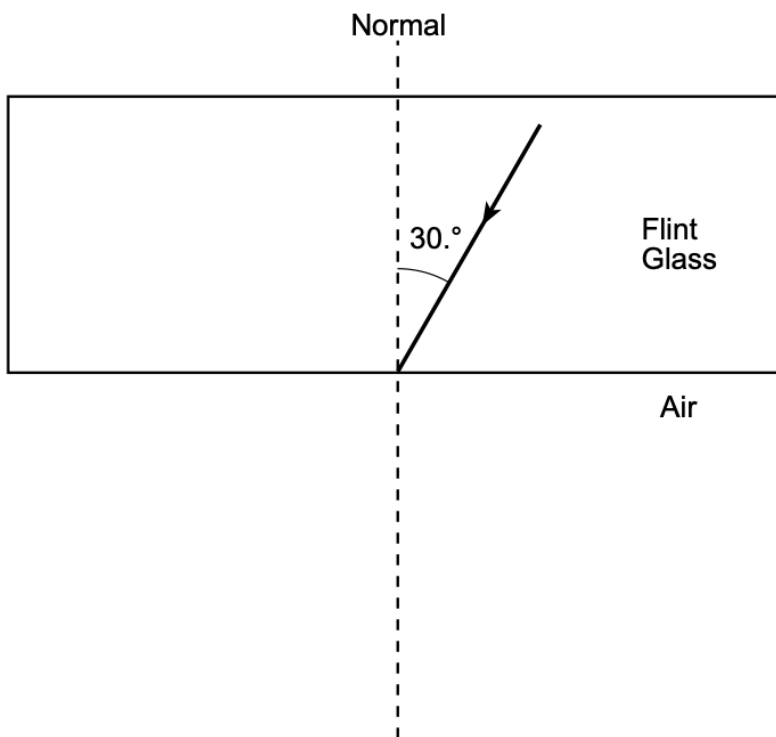
73–74. Calculate the child's kinetic energy at the bottom of the slide. *Show all work, including the equation and substitution with units.* [2]

75. Determine the total gravitational potential energy gained by the child when climbing 1.5 meters vertically from the bottom to the top of the slide. [1]

76. Determine the total amount of mechanical energy lost by the child–slide system as the child slides from the top to the bottom of the slide. [1]

Base your answer to questions 77 through 80 on the information and diagram below and on your knowledge of physics.

The diagram below represents a ray of light ($f = 5.09 \times 10^{14}$ Hz) in flint glass incident on air at an angle of incidence of 30° .



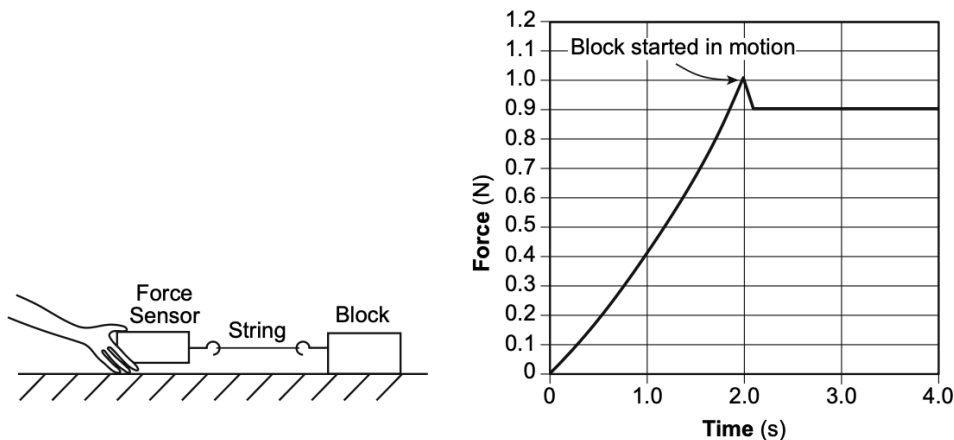
77–78. Calculate the angle of refraction of the light ray in the air. *Show all work, including the equation and substitution with units.* [2]

79. Using a protractor and a straightedge, draw the refracted light ray on the diagram in your answer booklet. [1]

80. State what happens to the wavelength of the light ray as the light ray passes into the air from the flint glass. [1]

Base your answers to questions 81 through 85 on the information, graph, and diagram below and on your knowledge of physics.

A student pulls a block weighing 2.0 newtons, resting on a horizontal laboratory table, with a string and force sensor. The sensor measures the force applied to the block when the block is at rest and when the block is moving. The force applied to the block is plotted against time on the graph.



81. Based on the graph, determine the force of kinetic friction acting on the block as it is pulled at constant velocity across the laboratory table. [1]

82–83. Calculate the coefficient of kinetic friction between the block and the laboratory table. *Show all work, including the equation and substitution with units.* [2]

84. At the instant the block begins to move, what happens to the value of the coefficient of friction? [1]

85. How would the coefficient of kinetic friction (μ) between the block and the laboratory table be affected if a 0.50-kilogram mass were placed on the block? [1]

Part C

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68. Determine the direction of the car’s velocity after the 6.0 seconds. [1]

69–70. Calculate the total distance traveled by the car during the 6.0 seconds. *Show all work, including the equation and substitution with units.* [2]

66–67 Take west as the negative direction. Using $v_f = v_i + at$:

$$v_i = -25 \text{ m/s}, \quad a = +1.5 \text{ m/s}^2, \quad t = 6.0 \text{ s},$$

$$\begin{aligned} v_f &= (-25 \text{ m/s}) + (+1.5 \text{ m/s}^2)(6.0 \text{ s}) \\ &= -25 \text{ m/s} + 9.0 \text{ m/s} = -16 \text{ m/s}. \end{aligned}$$

$$\boxed{|v_f| = 16 \text{ m/s}}$$

68 The final velocity is negative (-16 m/s), so the car is still moving westward.

69–70 For uniform acceleration:

$$\Delta x = v_i t + \frac{1}{2} a t^2.$$

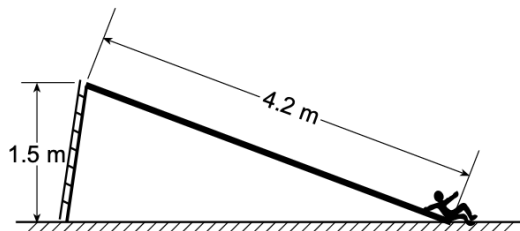
$$\begin{aligned} \Delta x &= (-25 \text{ m/s})(6.0 \text{ s}) + \frac{1}{2} (+1.5 \text{ m/s}^2)(6.0 \text{ s})^2 \\ &= -150 \text{ m} + 0.75(36) \text{ m} \\ &= -150 \text{ m} + 27 \text{ m} = -123 \text{ m}. \end{aligned}$$

Hence the car travels 123 m (magnitude):

$$\boxed{d = 1.23 \times 10^2 \text{ m}}$$

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71–72. Calculate the speed of the child at the instant she reaches the bottom of the slide. *Show all work, including the equation and substitution with units.* [2]

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71–72 Uniform-acceleration formula:

$$v^2 = v_0^2 + 2as$$

$$v^2 = 0 + 2(0.30 \text{ m/s}^2)(4.2 \text{ m}) = 2.52 \text{ m}^2/\text{s}^2$$

$$v = \sqrt{2.52} \text{ m/s} \approx 1.6 \text{ m/s}$$

$v \approx 1.6 \text{ m/s}$

73–74

$$K = \frac{1}{2}mv^2 = \frac{1}{2}(40 \text{ kg})(1.6 \text{ m/s})^2 \approx 50 \text{ J}$$

$K \approx 5.0 \times 10^1 \text{ J}$

75

$$\Delta U = mgh = (40 \text{ kg})(9.8 \text{ m/s}^2)(1.5 \text{ m}) \approx 5.9 \times 10^2 \text{ J}$$

$\Delta U \approx 5.9 \times 10^2 \text{ J}$

76 Total mechanical energy at the top: $E_{\text{top}} = \Delta U$ (KE is zero).

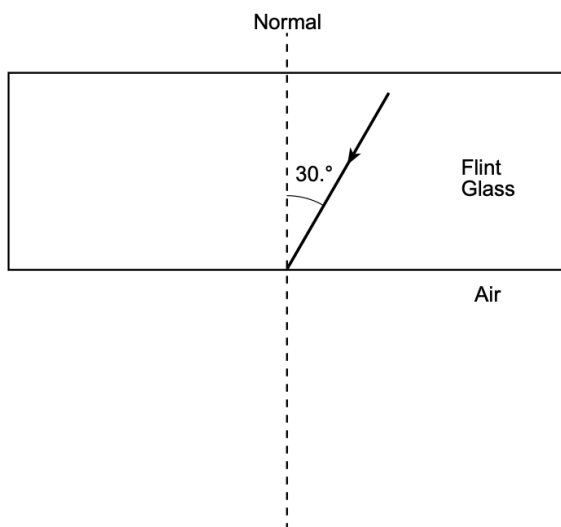
Total mechanical energy at the bottom: $E_{\text{bottom}} = K$.

$$\text{Energy lost} = E_{\text{top}} - E_{\text{bottom}} \approx (5.9 \times 10^2 \text{ J}) - (5.0 \times 10^1 \text{ J}) \approx 5.4 \times 10^2 \text{ J}$$

$\text{Energy lost} \approx 5.4 \times 10^2 \text{ J}$

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80. State what happens to the wavelength of the light ray as the light ray passes into the air from the flint glass. [1]

77–78 Snell's law: $n_{\text{glass}} \sin \theta_i = n_{\text{air}} \sin \theta_r$.

$$n_{\text{glass}} = 1.66, \quad n_{\text{air}} = 1.00, \quad \theta_i = 30^\circ$$

$$\sin \theta_r = \frac{n_{\text{glass}}}{n_{\text{air}}} \sin \theta_i = 1.66 (0.500) = 0.83$$

$$\theta_r = \sin^{-1}(0.83) \approx 5.6 \times 10^1^\circ$$

$$\theta_r \approx 56^\circ$$

79 With a protractor, measure an angle of approximately 56° on the air side of the normal (bending away from the normal) and draw the refracted ray as a straight line at that angle. (The ray should emerge above the original interface, making a wider angle than the incident ray.)

80 The frequency remains constant, but the speed of light increases when it enters air; therefore the wavelength increases as the ray passes from the flint glass into the air.

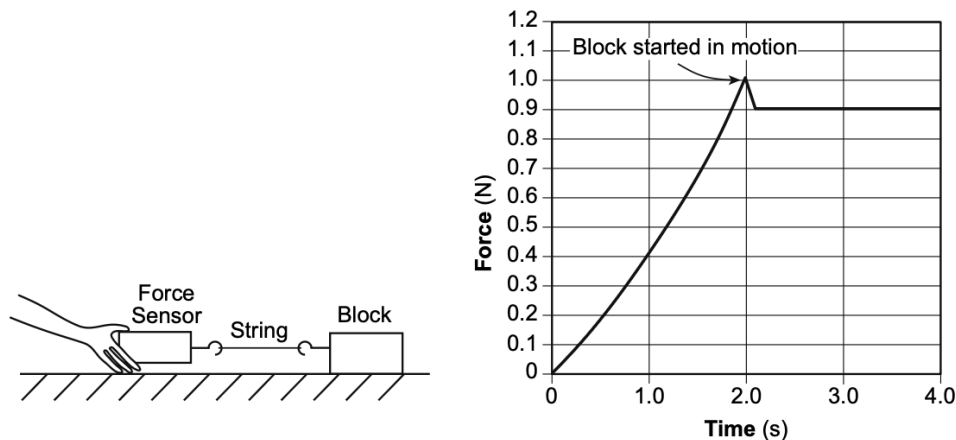
$$\lambda_{\text{air}} > \lambda_{\text{glass}}$$

The oscillation frequency f is set by the source and therefore cannot change at the boundary. Since the index of refraction drops from n_{glass} to n_{air} , the wave speed rises $v = c/n$ when the light enters air. With $v = f\lambda$ and f constant, the wavelength must stretch:

$$\lambda_{\text{air}} = \frac{v_{\text{air}}}{f} = \frac{c/n_{\text{air}}}{f} > \frac{c/n_{\text{glass}}}{f} = \lambda_{\text{glass}}$$

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81 The flat portion of the graph (constant-velocity phase) is at about 0.90 N.

$$F_k \approx 0.90 \text{ N}$$

82–83

$$\mu_k = \frac{F_k}{N} = \frac{0.90 \text{ N}}{2.0 \text{ N}} = 0.45$$

$$\mu_k \approx 0.45$$

84 When the block just begins to slide, the static-friction coefficient is replaced by the lower kinetic coefficient; therefore the coefficient of friction decreases.

85 Adding mass increases the normal force and the kinetic-friction force by the same factor, but the ratio $\mu_k = F_k/N$ is unchanged.

No change in μ_k